



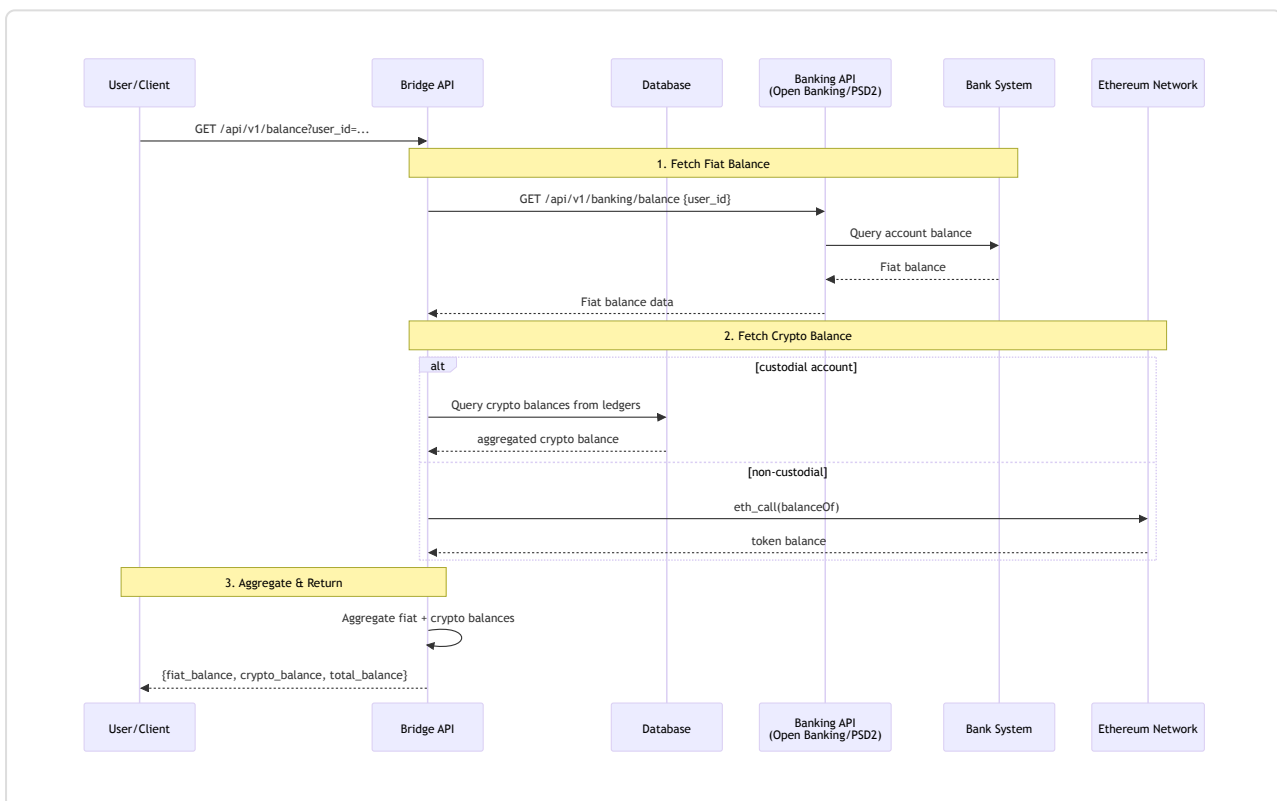
BRIDGE-API Flow Diagrams

Overview: These diagrams illustrate the complete banking-to-crypto bridge flows, including proper banking integration layers, compliance checks, and event-driven processing between traditional banking systems and blockchain networks.



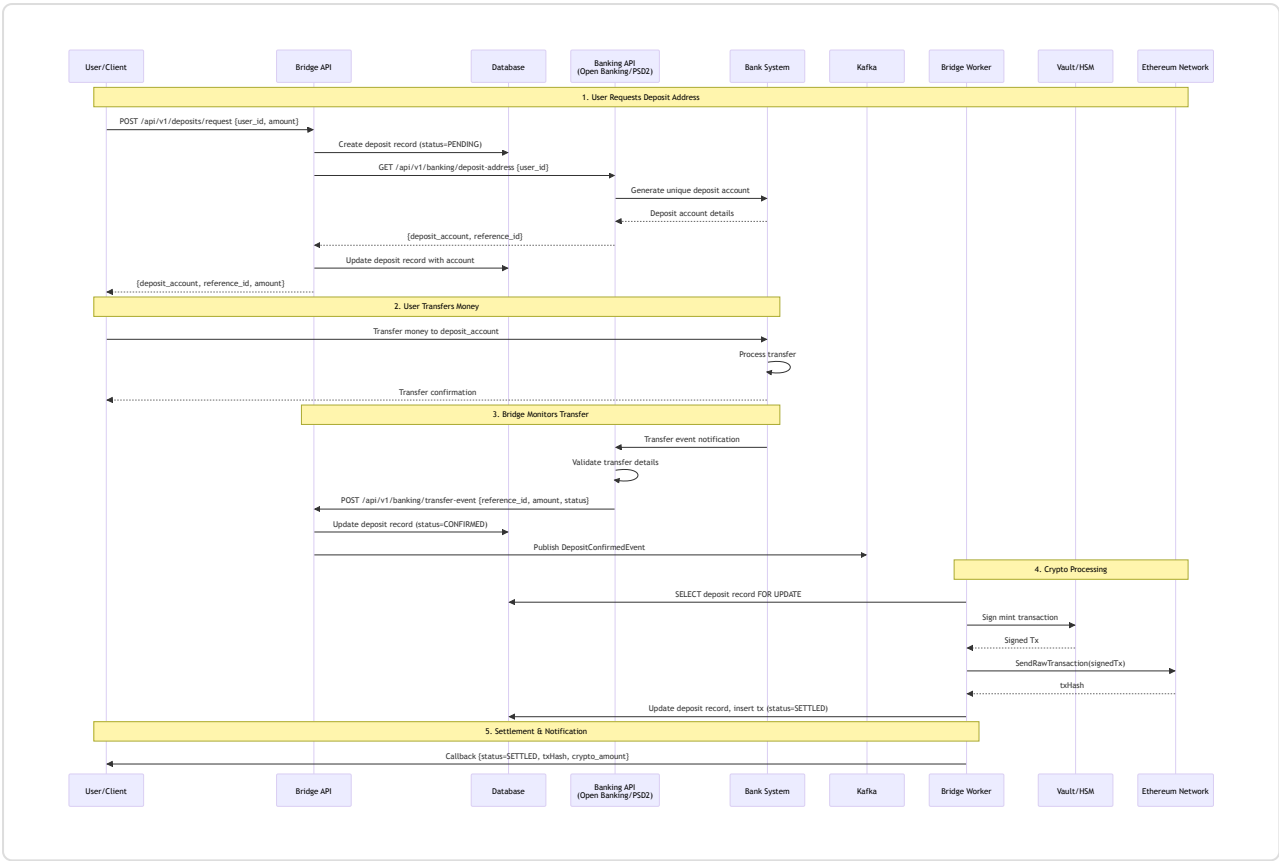
Balance Query Flow

Purpose: Shows how users can query their complete account balances, including both fiat (from banking system) and crypto (from blockchain) balances with proper banking integration.



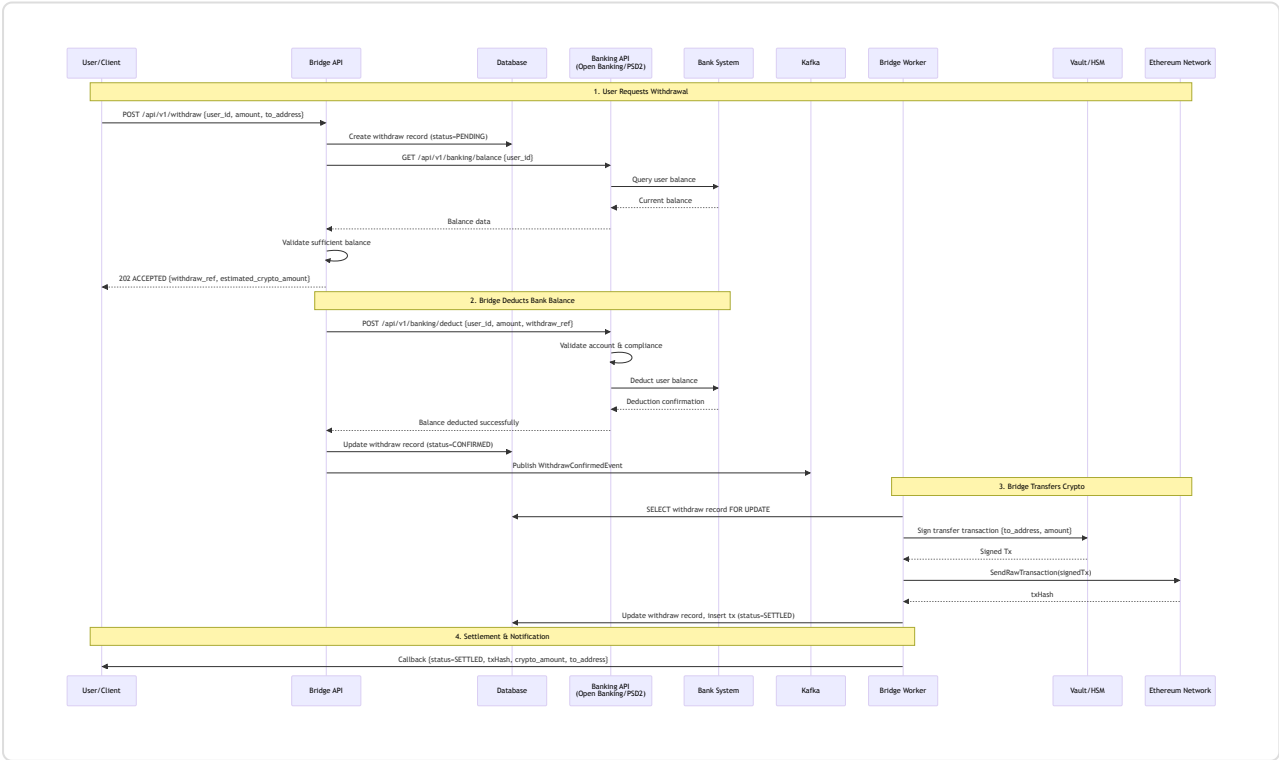
Deposit Flow

Purpose: Illustrates the user-initiated deposit process where users request deposit addresses, transfer money to those addresses, and receive crypto credits through event-driven processing.



Withdraw Flow

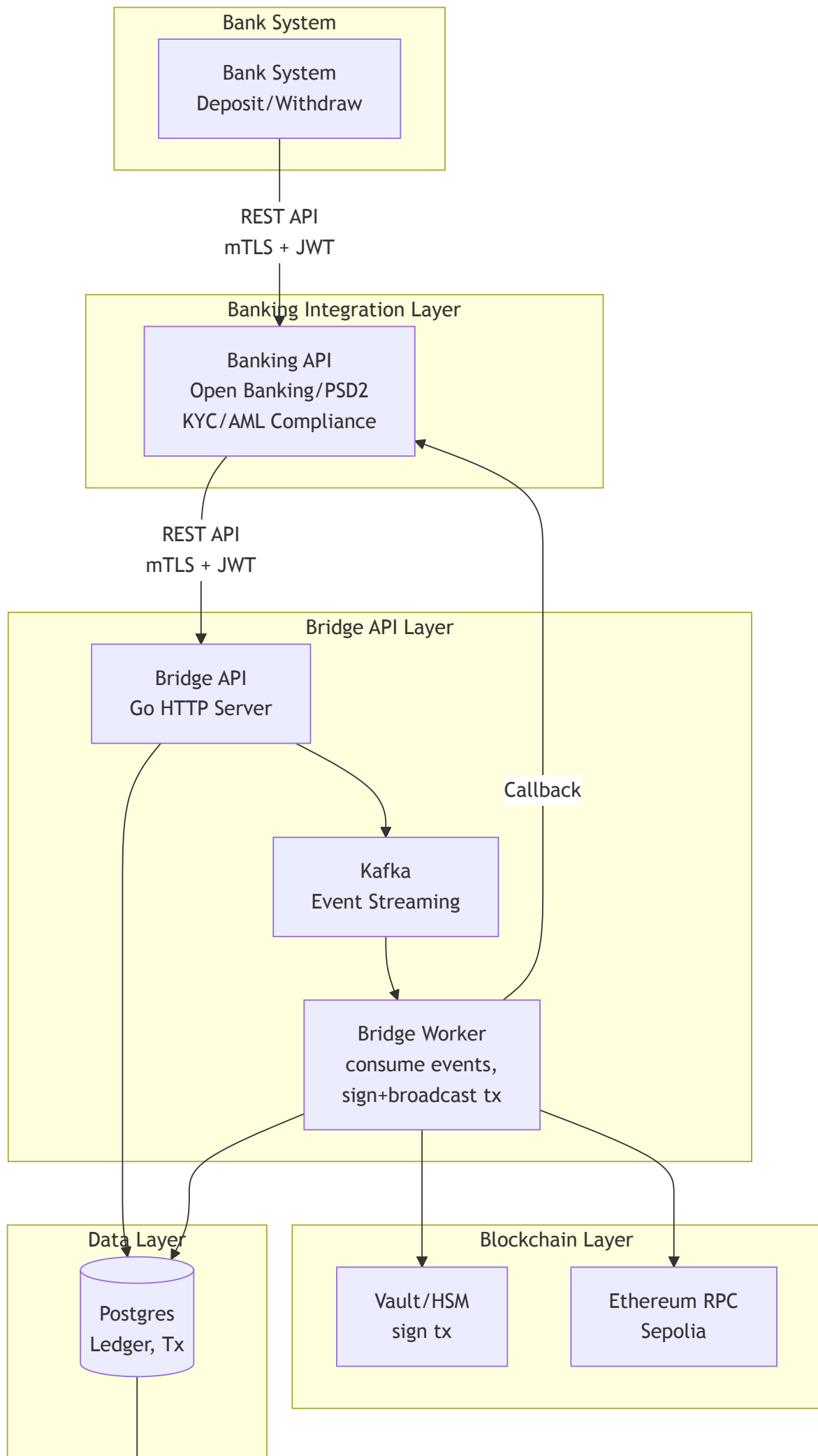
Purpose: Shows the user-initiated withdrawal process where users provide blockchain addresses, bridge deducts bank balance, and transfers crypto to the specified address.

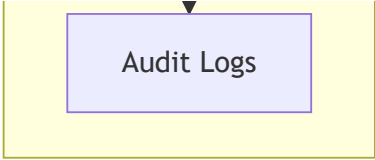




System Architecture

Purpose: High-level overview of the BRIDGE-API system components and their interactions, including the banking integration layer.





```
graph TD; subgraph System; direction TB; AL[Audit Logs]; end;
```

Audit Logs

Key Components:

- **Banking API:** Open Banking/PSD2 integration with KYC/AML compliance
- **Bridge API:** Main HTTP server handling REST API requests
- **Kafka:** Event streaming for asynchronous processing
- **Worker:** Background service processing blockchain transactions
- **Vault/HSM:** Secure key management for transaction signing
- **PostgreSQL:** Ledger and transaction storage
- **Ethereum:** Blockchain network (Sepolia testnet)



Security & Compliance Features

Purpose: Overview of the security and compliance features built into the banking-to-crypto bridge system.

